

27/02/26

Exp. no : 3

Shell Script

Aim:

To study and execute all the commands in shell script and verify it.

To run the script:

```
sh basic-shell-demo.sh
```

Shell Programming:

1. Output Using "ECHO":

```
echo "Hello, this is my first shell script"
```

2. Variables:

```
NAME = "Bam"
```

```
AGE = "22"
```

```
echo "Name is $NAME"
```

```
echo "Age is $AGE"
```

3. Constant Variables:

```
readonly COUNTRY = "India"
```

```
COUNTRY = "NEPAL"
```

```
echo "Country is $COUNTRY"
```

4. Read input from User:

```
echo "Enter your name: "
```

```
read username
```

```
echo "Hello $username"
```

5. IF-ELSE Statement:

```
usage = 10
```

```
if [ $usage -ge 18 ]
```

```
then
```

```
echo "You are eligible"
```

```
else
```

```
echo "You are not eligible"
```

```
fi
```

6. If - Elif - Else :

COUNTRY = "Nepal"

if ["\$COUNTRY" = "India"]

then

echo " You are from India "

elif ["\$COUNTRY" = "Nepal"]

then

echo " You are from Nepal "

else

echo " Country not matched "

fi

Operators :

Arithmetic : + - * / %

Relational :

- eq (equal)

- ne (not equal)

- gt (greater than)

- lt (less than)

- ge (greater than equal)

- le (less than equal)

1. For Loop :

n=10

for ((i=1 ; i<=n ; i++))

do

echo \$i

done

8. While Loop

```
count=1
num=5
while [ $count -le $num ]
do
    echo "Count is $count"
    count=$((count+1))
done
```

9. Break and continue:

```
for i in 1 2 3 4 5
do
    if [ $i -eq 3 ]
    then
        continue
    fi
    if [ $i -eq 5 ]
    then
        break
    fi
    echo "Value: $i"
done
```

10. Expression:

```
a=10
b=5
sum=a+b
let sum=a+b
mul=$((a*b))
a='expr $a + 1'
echo "Sum is $sum"
echo "Multiplication is $mul"
echo "Value of 'a' is $a"
```


11. Functions:

```
myfun() {  
    echo "Hello from function"  
}  
# functional call  
myfun
```

12. Function with Arguments:-

```
add() {  
    echo "First value : $1"  
    echo "Second value : $2"  
    echo "Sum : $(( $1 + $2 ))"  
}
```

13. Script Arguments:

```
echo "Script name : $0"  
echo "First argument : $1"  
echo "Second argument : $2"  
echo "Total arguments : $#"  
echo "All arguments : $@"
```

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Program Using Shell Script

1. Arithmetic operations

```
#!/bin/bash
echo "Enter two Numbers"
read a
read b

c = "expr $a + $b"
d = "expr $a - $b"
e = "expr $a * $b"
f = "expr $a / $b"
g = "expr $a % $b"

echo "add $c"
echo "sub $d"
echo "mul $e"
echo "div $f"
echo "mod $g"
```

Output:

```
Enter two numbers: 10 5
add 15
sub 5
mul 50
div 2
mod 0
```


2. Leap Year

```
#!/bin/bash
echo "Enter number"
read a
m = `expr $a % 4`
if [ $m -eq 0 ]
then
    echo "Leap Year"
else
    echo "Not a leap year"
fi
```

Output:

```
Enter number
2024
Leap year
```

3. Reverse Number

```
#!/bin/bash
echo "Enter number"
read n
rev = 0
while [ $n -gt 0 ]
do
    d = $((n % 10))
    rev = $((rev * 10 + d))
    n = $((n / 10))
done
echo "Reversed number is $rev"
```

Output:

Enter number

416

Reversed number is 614

4. Fibonacci Series

```
#!/bin/bash
echo "Enter number"
read n
a=0
b=1
echo "Fibonacci series:"
for ((i=1; i<n; i++))
do
    echo $a
    c=$((a+b))
    a=$b
    b=$c
done
```

Output:

Enter number

5

Fibonacci series

~~0 1 1 2 3~~
0
1
1
2
3

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Result:

Thus the shell script commands has been executed and verified.